EE2001 Electrical Circuits- I:

The Operational Amplifier

6 November 2023

Laboratory Announcement

1

- **Experiment 2** will be held this week (Week 6) Thursday (9 November)-Friday (10 November) and the next week (Week 7) Thursday (16 November)-Friday (17 November).
- You can only enter the experiment on the single specific day scheduled for your group. Please check the lists and find which day you and your group will enter the experiment:

9 NOVEMBER THURSDAY GROUPS LIST:

https://drive.google.com/file/d/1GLmfi25NUvNYFXMR_oALO4R6KXBfC0hd/view?usp=drive_link

10 NOVEMBER FRIDAY GROUPS LIST:

https://drive.google.com/file/d/1RteEIJ3GWuQGVzDIanG-ow67dQPax4IS/view?usp=drive_link

16 NOVEMBER THURSDAY GROUPS LIST:

https://drive.google.com/file/d/17uM247YSaLHf1E70969Lu35-v6N_ySGe/view?usp=drive_link

17 NOVEMBER FRIDAY GROUPS LIST:

https://drive.google.com/file/d/1NQKsodaqZDi5anz5sMUNeaEaoBgmuiSU/view?usp=drive_link

Outline



1. Introduction
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3. Terminal Voltages and Currents
4. The Inverting-Amplifier Circuit
5. The Summing-Amplifier Circuit
6. The Noninverting-Amplifier Circuit
7. The Difference-Amplifier Circuit

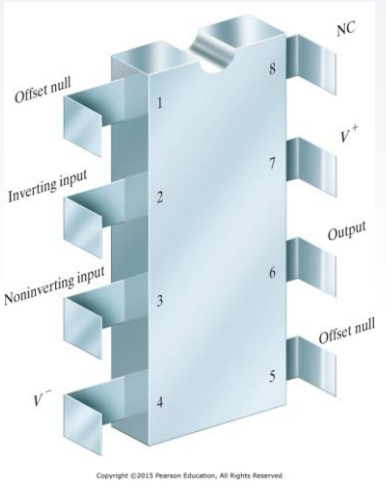
1 Introduction

- The Operational Amplifier (**Op-Amp**) is a widely used electronic device.
- A **detailed** analysis of operational amplifier circuit **requires understanding of electronic devices** such as diodes and transistors.
- However, in this course, the circuit is introduced before discussing the circuit's components, because,
  - To show you **how it can be used as a circuit building block** by focusing on its **terminal behavior** (**black box** approach),
  - You can **combine it with resistors** to perform some very **useful functions,** such as **scaling, summing, sign changing and subtracting**.

Operational Amplifier Terminals

2

- A **commercially available** operational amplifier device is **μA741**.
- The device has **eight terminals** (eight-lead Dual in-line package **DIP**).
  - NC = **Not connected**
  - **Offset null terminals** can be used to connect external circuitry to compensate for degradation of performance due to aging etc. However, **usually** degradation is negligible and these terminals are **not used**.
- The five important terminals are,
  - **Inverting** input (-)
  - **Noninverting** input (+)
  - **Output**
  - **Positive** power supply (V^+)
  - **Negative** power supply (V^-)

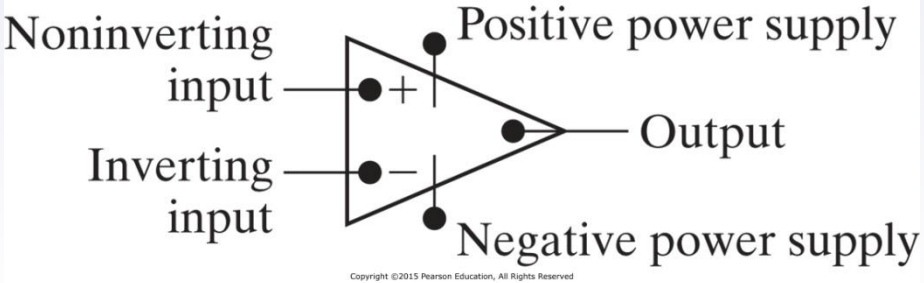


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Operational Amplifier Terminals

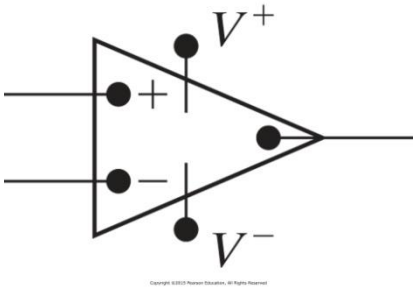
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- The **circuit symbol** of the operational amplifier is



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- **For sim**plicity, in circuits the op-amp is drawn as,

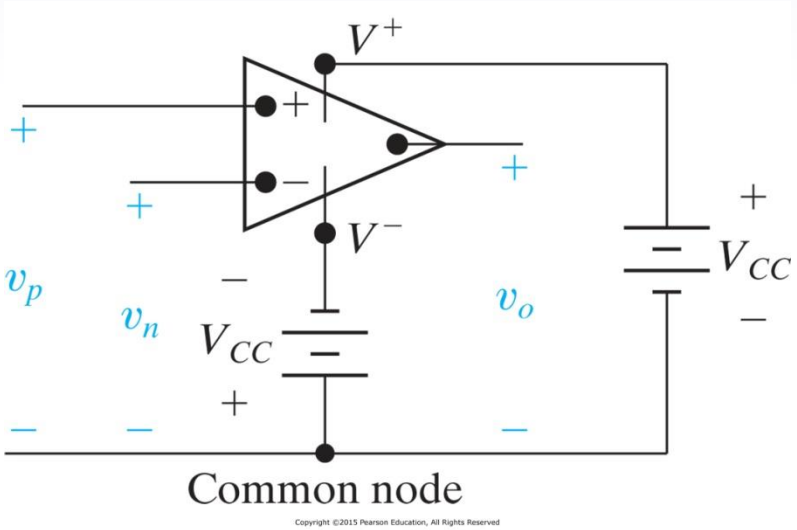


- Or **even sometimes** the V^+ **and** V^- **terminals are ommitted** (**keeping in mind** that **<u>they are connected</u>**).
- In order for the op-amp **to be operational**, first the **supply voltages need to be connected**.

Terminal Voltages and Currents

- The suppy voltage and all the other voltages are **measured with respect to a common reference** node (which **is usually the ground**).

- v_p = the voltage at the noninverting input
- v_n = the voltage at the inverting input
- v_o = the output voltage
- V_{CC} = the positive supply voltage
- $-V_{CC}$= the negative supply voltage
- All these voltage are **with respect to the common node** (which is **usually the ground**).

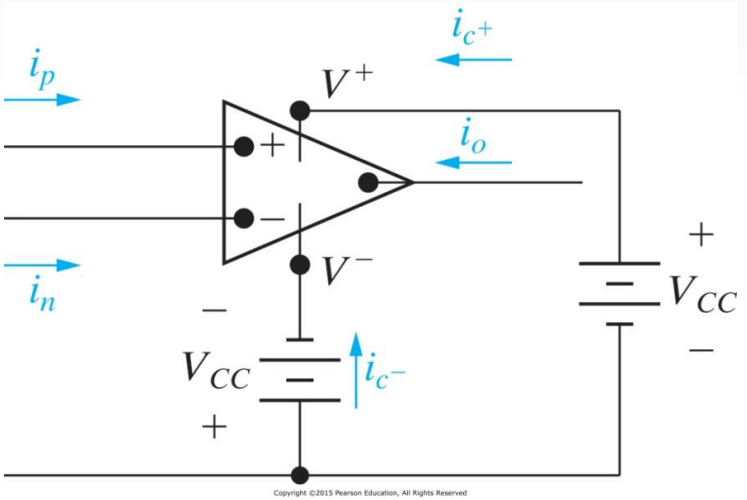


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3 Terminal Voltages and Currents

- We can consider the **currents through the terminals** of the device.

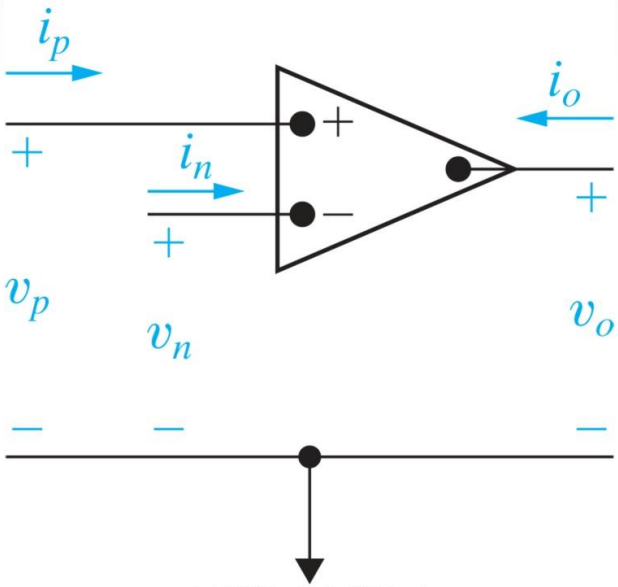
- i_p = the current through the noninverting input
- i_n = the current through the inverting input
- i_o = the current through the output
- i_{C+} = the current through the positive supply terminal
- i_{C-} = the current through the negative supply terminal
- The **reference direction** of all these currents are **towards the device**.



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3 Terminal Voltages and Currents

- Sometimes **for further simplicity**, the **supply voltages** and the **related terminals** are also **omitted** from the circuit **keeping in mind that they are connected (they are just not drawn)**.



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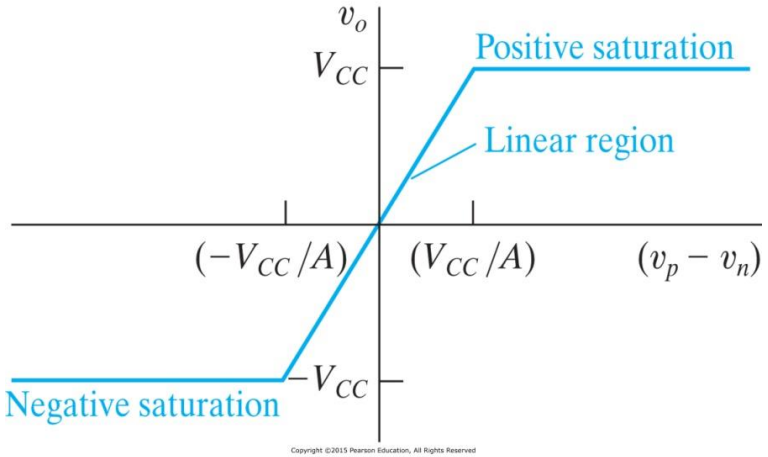
Terminal Voltages and Currents

3

- The **input-output voltage characteristics** of op-amp can be described as follows:

$$V_o = \begin{cases} -V_{CC}, & A(v_p - v_n) < -V_{CC} \\ A(v_p - v_n), & -V_{CC} < A(v_p - v_n) < +V_{CC} \\ +V_{CC}, & A(v_p - v_n) > +V_{CC} \end{cases}$$

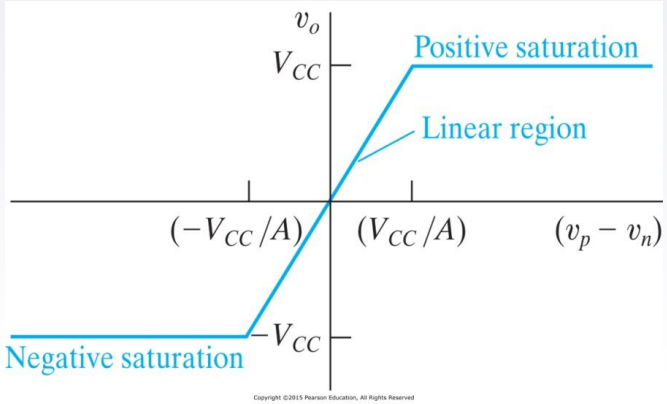
- This can be **represented in graphical** form as:



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Terminal Voltages and Currents

$$V_o = \begin{cases} -V_{CC}, & A(v_p - v_n) < -V_{CC} \\ A(v_p - v_n), & -V_{CC} < A(v_p - v_n) < +V_{CC} \\ +V_{CC}, & A(v_p - v_n) > +V_{CC} \end{cases}$$



- The **constant A** is called the **gain** of the op-amp.
- The values of the power supply voltages $V^+ = \boldsymbol{V_{CC}}$ and $V^- = \boldsymbol{-V_{CC}}$
- When $|\boldsymbol{v_p - v_n}|$ **is small**, the op-amp **operates in a linear operation mode**.
- The **gain** of the op-amp **is usually large**.
- For $V_{CC} = 20V$ and $\boldsymbol{A = 10000 = 10^4}$ one needs

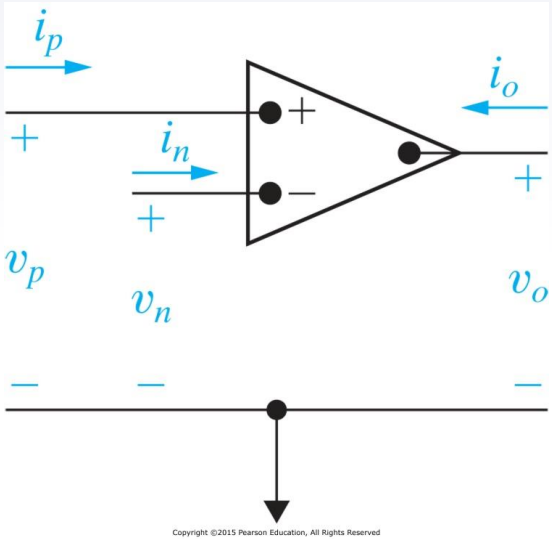
$$|v_p - v_n| < \frac{V_{CC}}{\mathrm{A}} = \frac{20}{10000} = 2\ \mathrm{mV}$$

in order to have the amplifier **operating in its linear region**.

Terminal Voltages and Currents

3

- In order to **simplify** the analysis, usually an **ideal op-amp model is assumed,** which satisfies,



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- $v_p = v_n$
- $i_p = i_n = 0$
- $A = \infty$
- Input resistance $= \infty$

- The input resistance of an op-amp is **usually very large** (e.g., **1 MΩ**) justifying the **current constraint condition** ($i_p = i_n = 0$). The **voltage constraint condition** $v_p = v_n$ is called the **virtual short condition**.

Terminal Voltages and Currents

3

- When an op-amp is connected in a circuit **in order to maintain** the virtual short condition, **a signal is fed back from the output to the inverting input**. This connection is called a **negative feedback** since the signal fed back from the output is subtracted from the input signal.
- A negative feedback causes the input **voltage diffe**rence between v_p and v_n **to decrease**.
- **Without such a feedback connection** the op-amp would **normally saturate**.
- The difference between the input signals **must be extremely small** in order to prevent saturation without a negative feedback.
- **Even when** there is negative feedback **saturation can still occur**.

Terminal Voltages and Currents

- **We normally do not know** whether the op-amp is **in linear mode or saturated**.
- Usually we **assume linear operation** and **analyze** the circuit.
- If during analysis **a contradiction is en**countered such as the **output voltage being larger than the supply voltage** one understands that the linear operation **assumption does not hold** and the op-amp **is saturated**.
- Expressing **the current equation** for an op-amp one obtains

$$i_p + i_n + i_o + i_{C+} + i_{C-} = 0$$

- For an **ideal op-amp** we have $i_p = i_n = 0$ implying that

$$i_o = -(i_{C+} + i_{C-})$$

- **Usually** i_{C+} and i_{C-} are **omitted** (not shown) in circuits.
- Therefore, one may thing that the output current is zero. However, this may be **misleading**.
- Note that even though the input currents of an op-amp are zero, there may be output current in an op-amp.

3 Terminal Voltages and Currents

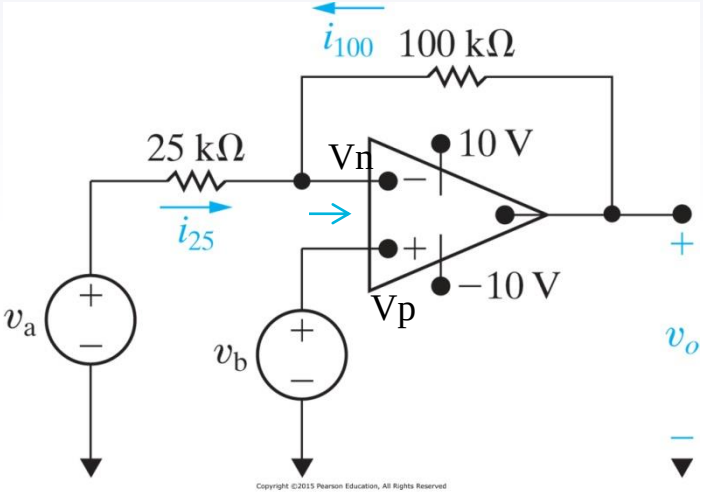
- Note also that the supply voltages V^+ and V^- **do not have to be equal**.
- The output voltage **has to lie between** the two supply voltages.
- In other words, $V^- \leq v_o \leq V^+$ **must be satisfied**.
  - For example, if $V^+ = 20\ V$ and $V^- = -15\ V$the output voltage must satisfy:

    -15 V$\leq v_o \leq 20\ V$.
- Moreover, the value of the **gain *A* is not constant under all operating conditions** and **may change**.

Terminal Voltages and Currents

3

- Example: The op-amp in the circuit is ideal.



a) Calculate v_o, if $v_a = 1\ V$ and $v_b = 0\ V$

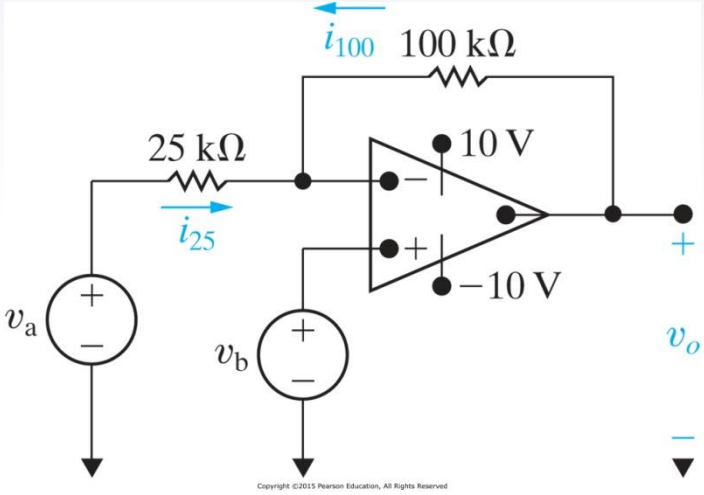
- Because a **negative feedback path exists** from the op-amp output to its inverting input through the 100 kΩ resistor, let's **assume** the op-amp **is confined to its linear operating region**.
- We can write **a node-voltage equation** at **the inverting input terminal**.
- The voltage at the **inverting input terminal is 0**, as $v_p = v_b = 0$ from **the connected voltage source**, and $v_n = v_p$ from **the voltage constraint**.
- Considering these, the **node voltage equation at v_n**:

$$i_{25} + i_{100} = i_n$$

Terminal Voltages and Currents

3

- Example: The op-amp in the circuit is ideal.



- From **Ohm's law**,

$$i_{25} = \frac{(v_a - v_n)}{25} = \frac{1}{25}\ mA$$

$$i_{100} = \frac{(v_o - v_n)}{100} = \frac{v_o}{100}\ mA$$

- The **current constraint** requires $i_n = 0$.
- **Substituting** the values for the three currents in the node voltage equation, we obtain,

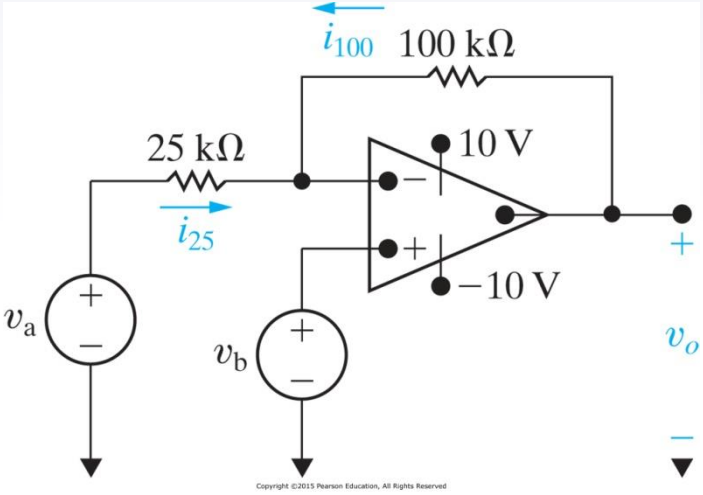
$$\frac{1}{25} + \frac{v_o}{100} = 0$$

- Hence, $v_o = -4V$. Note that because v_o **lies between** $\pm\ 10V$, the op-amp is **in its linear region of operation**.

Terminal Voltages and Currents

3

- Example: The op-amp in the circuit is ideal.



b) Repeat (a) for $v_a = 1\ V$ and $v_b = 2V$

- Using the **same process as in (a)**, we get:

$$v_p = v_b = v_n = 2V$$

$$i_{25} = \frac{(v_a - v_n)}{25} = \frac{(1-2)}{25} = -\frac{1}{25}\ mA$$

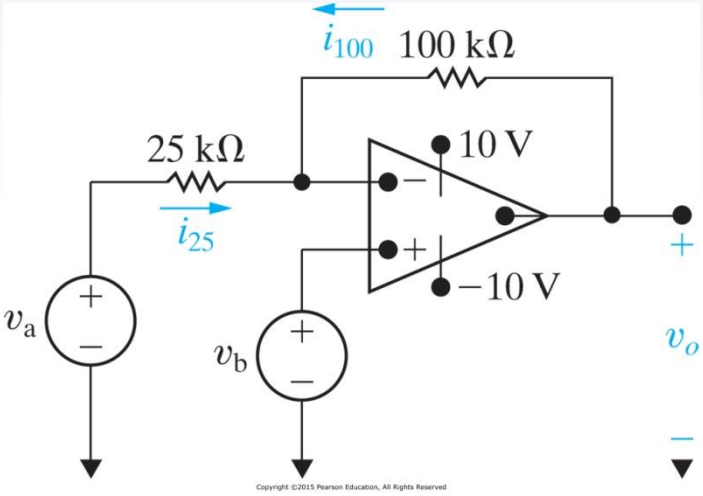
$$i_{100} = \frac{(v_o - v_n)}{100} = \frac{v_o - 2}{100}\ mA$$

$$i_{25} = -i_{100}$$

- Therefore, $v_o = 6V$. Again v_o **lies within** $\pm 10V$, .

Terminal Voltages and Currents

- Example: The op-amp in the circuit is ideal.



c) If $v_a = 1.5\ V$ specify the range of v_b **that avoids amplifier saturation**.

- As before, $v_n = v_p = v_b$ and $i_{25} = -i_{100}$. Because $v_a = 1.5V$

$$\frac{1.5 - v_b}{25} = -\frac{(v_o - v_b)}{100}$$

- **Solving** for v_b as a function of v_o gives:

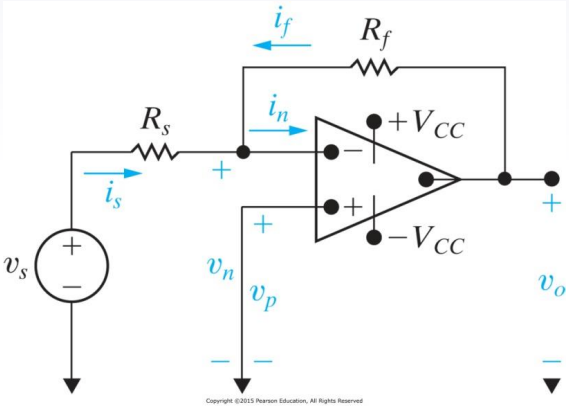
$$v_b = \frac{1}{5}(6 + v_o) \qquad 5v_b - 6 = v_o$$

- Now, if the amplifier is **to be within the linear region of operation**, -10V$\leq v_o \leq 10V$.
- **Substituting** these limits on v_o into the expression for v_b is limited to,

$$\text{-0.8V} \leq v_b \leq 3.2V$$

The Inverting-Amplifier Circuit

- Operational amplifiers can be used to **achieve various useful circuits**.
- The inverting amplifier circuit **is one of such circuits**.
- Consider the following circuit with the op-amp **assumed to be ideal**.



$$i_p = i_n = 0 \qquad v_p = v_n = 0$$

$$i_s = \frac{v_s}{R_s} \qquad i_f = \frac{v_o}{R_f}$$

$$i_s = -i_f$$

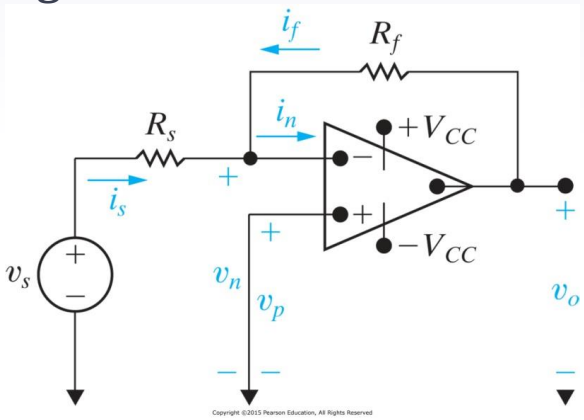
$$\frac{v_s}{R_s} = -\frac{v_o}{R_f} \qquad v_o = -\frac{R_f}{R_s} v_s$$

- The **output voltage** is the **scaled inverted input**.
- $\boldsymbol{R_f/R_s}$ is the **amplifier gain** (**scaling factor**).
- This **holds for an ideal op-amp**. For a practical op-amp it is an approximation (usually a sufficiently good one).

The Inverting-Amplifier Circuit

4

- Given the op-amp **gain A is large enough**, the inverting-amplifier gain can be **specified by the external resistors R_f and R_s.**
- In order for the discussed amplification to hold, the op-amp must be **operating in its linear region.**



$$V^- \le v_o \le V^+ \qquad V^- = -V_{CC} \qquad V^+ = +V_{CC}$$

$$-V_{CC} \le -\frac{R_f}{R_s} v_s \le +V_{CC} \qquad \left|-\frac{R_f}{R_s} v_s\right| \le V_{CC}$$

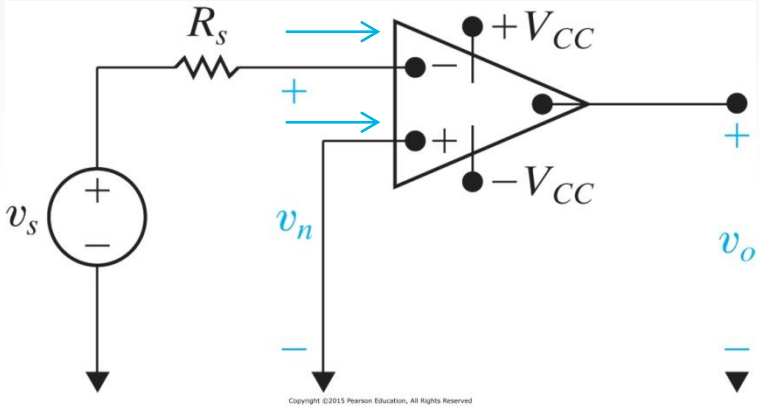
$$\frac{R_f}{R_s} \le \left|\frac{V_{CC}}{v_s}\right|$$

- For the op-amp to be opearting in its linear region, the **above condition must be satisfied**.
- In the above circuit the resistor **R_f provides a negative feedback connection** (in order to prevent the op-amp from saturation).

The Inverting-Amplifier Circuit

4

- **If the resistor R_f is removed**, the feedback path is removed and the **amplifier is operating open loop**.



$$v_o = -Av_n \qquad V^- = -V_{CC} \qquad V^+ = +V_{CC}$$

$$|v_n| \leq \frac{V_{CC}}{A}$$

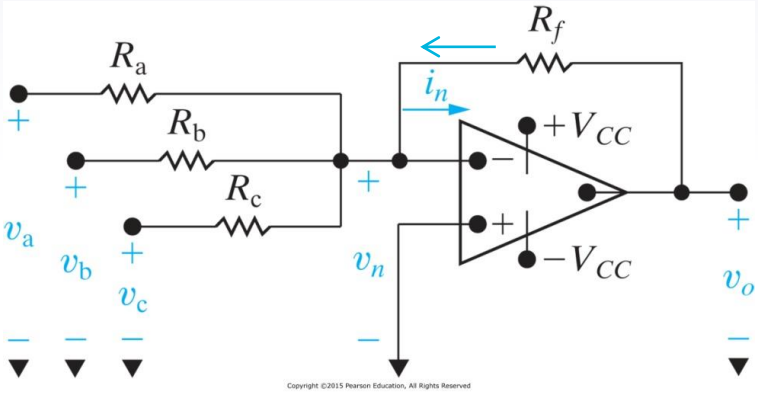
$$v_n \cong v_s \qquad |v_s| \leq \frac{V_{CC}}{A}$$

- **Otherwise**, the op-amp **saturates**.

The Summing-Amplifier Circuit

5

- If more than one input voltages are connected to the inverting terminal in the inverting amplifier, it becomes a summing amplifier.
- Consider the circuit with the op-amp assumed to be ideal.



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$$i_p = i_n = 0 \qquad v_p = v_n = 0$$

$$\frac{v_a}{R_a} + \frac{v_b}{R_b} + \frac{v_c}{R_c} + \frac{v_o}{R_f} = 0$$

$$v_o = -\left(\frac{R_f}{R_a} v_a + \frac{R_f}{R_b} v_b + \frac{R_f}{R_c} v_c\right)$$

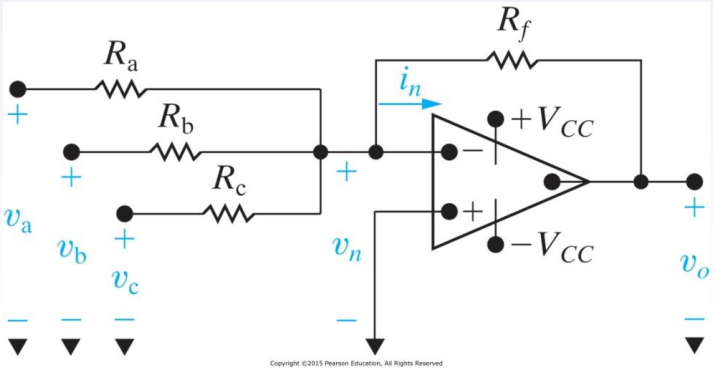
- The output voltage is the weighted and inverted sum of the input voltages.

The Summing-Amplifier Circuit

5

- In the summing amplifier circuit, if all the input resistors are chosen equal
- $(R_a = R_b = R_c = R_s)$, then

$$v_o = -\frac{R_f}{R_s}(v_a + v_b + v_c)$$



- Furthermore, if $R_f = R_s$, then

$$v_o = -(v_a + v_b + v_c)$$

- The number of input voltages can be changed as needed.
- The values of the input and feedback resistances can be changed to achieve desired scaling factors.
- For proper operation, the output voltage still needs to satisfy the condition:

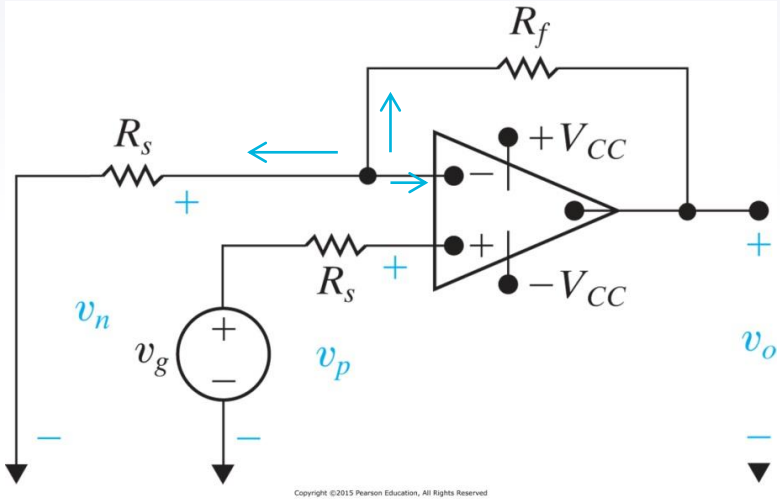
$$V^- \le v_o \le V^+ \qquad V^- = -V_{CC} \qquad V^+ = +V_{CC}$$

$$-V_{CC} \le \left(\frac{R_f}{R_a}v_a + \frac{R_f}{R_b}v_b + \frac{R_f}{R_c}v_c\right) \le +V_{CC}$$

The Noninverting-Amplifier Circuit

6

- Using an op-amp, it is possible to achieve amplification without inversion as well.
- Consider the circuit below with the op-amp assumed to be ideal.



$$i_p = i_n = 0 \qquad v_p = v_n$$

$$v_p = v_g \qquad v_n = v_g$$

$$v_n = v_g = \frac{v_o R_s}{R_s + R_f}$$

Also from $\frac{v_g}{R_s} + \frac{v_g - v_o}{R_f} = 0$ $\qquad \frac{v_g}{R_s} + \frac{v_g}{R_f} = \frac{v_o}{R_f}$

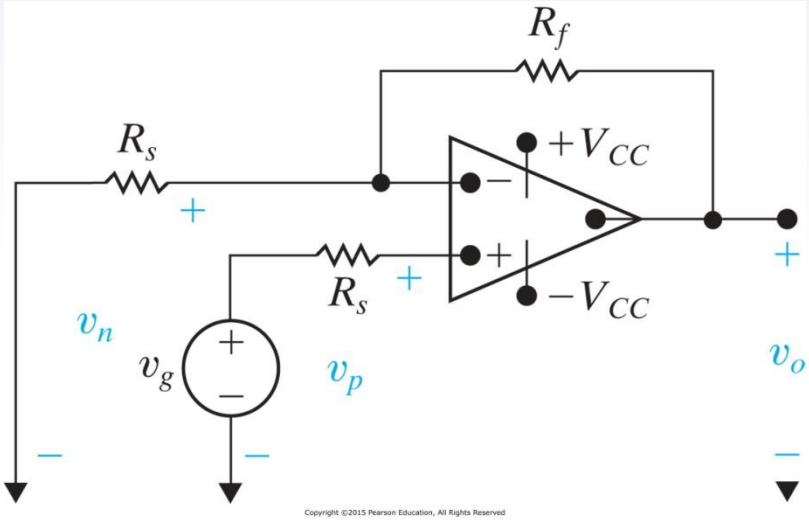
$$v_o = \left(\frac{R_s + R_f}{R_s}\right) v_g = \left(1 + \frac{R_f}{R_s}\right) v_g$$

- The output voltage is the scaled replica of the input voltage.
- $\left(1 + \frac{R_f}{R_s}\right)$ is the amplifier gain.

The Noninverting-Amplifier Circuit

6

- In order to have op-amp operating in its linear operation mode, the following must be satisfied.



$$V^- \le v_o \le V^+$$

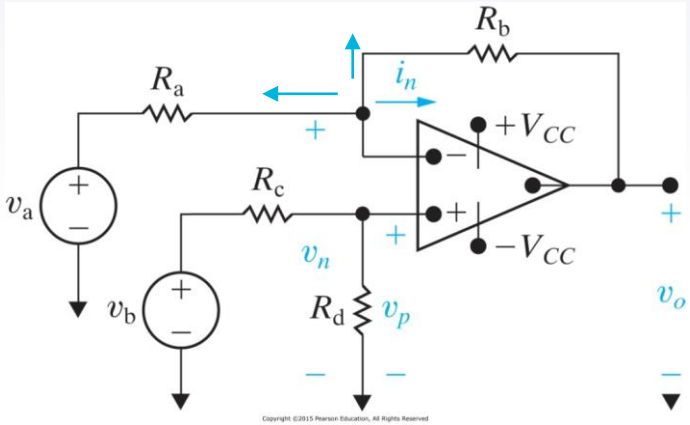
$$V^- = -V_{CC} \qquad V^+ = +V_{CC}$$

$$-V_{CC} \le \left(1 + \frac{R_f}{R_s}\right) v_g \le +V_{CC}$$

The Difference Amplifier Circuit

7

- The difference amplifier takes the difference between two input voltages.
- Consider the following circuit with the op-amp **assumed to be ideal**.



$$i_p = i_n = 0$$

$$v_p = v_n$$

$$v_p = \frac{R_d}{R_c + R_d} v_b$$

$$\frac{v_n - v_a}{R_a} + \frac{v_n - v_o}{R_b} = 0$$

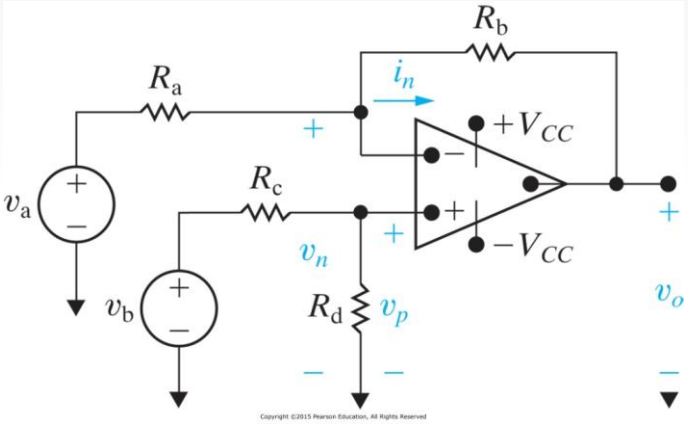
$$v_o = \left(1 + \frac{R_b}{R_a}\right) v_n - \frac{R_b}{R_a} v_a$$

$$v_o = \left(\frac{R_a + R_b}{R_a}\right)\left(\frac{R_d}{R_c + R_d}\right) v_b - \frac{R_b}{R_a} v_a$$

- The **output voltage** is the **weighted difference between the two input voltages.**

The Difference Amplifier Circuit

- The scaling factors (i.e., the weights) of the two voltages in the difference amplifier are in general not equal.
- It is possible to derive a condition for them to be equal.



$$\left(\frac{R_a + R_b}{R_a}\right)\left(\frac{R_d}{R_c + R_d}\right) = \frac{R_b}{R_a}$$

$$\left(1 + \frac{R_b}{R_a}\right) = \frac{R_b}{R_a}\left(1 + \frac{R_c}{R_d}\right)$$

$$1 = \frac{R_b}{R_a}\frac{R_c}{R_d}$$

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

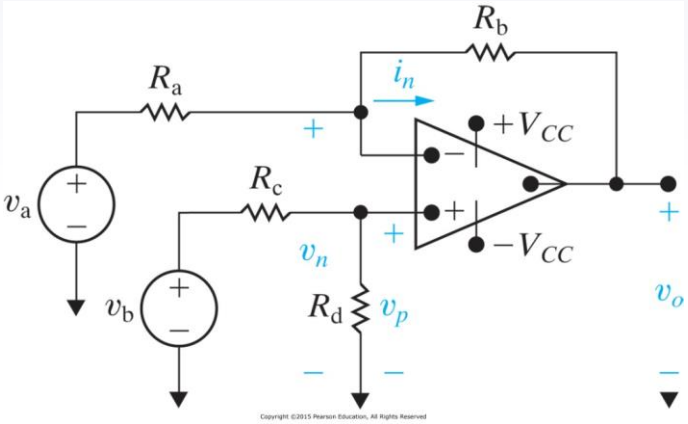
$$v_o = \frac{R_b}{R_a}(v_b - v_a)$$

- If the condition here is satisfied, the output voltage is the scaled difference of the input voltages.
- R_b/R_a is the gain of the difference amplifier.

The Difference Amplifier Circuit

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- In addition to the condition on the previous page if $R_b = R_a$ in the difference amplifier, then the output voltage is expressed as just the difference between the input voltages.



$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

$$R_b = R_a$$

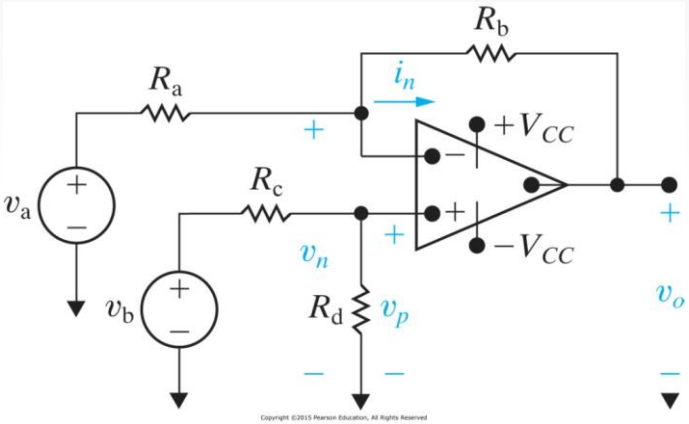
$$v_o = v_b - v_a$$

- As was in all previous amplifiers the scaling of the input voltages in the difference amplifier are determined by the resistors and can be adjusted by proper choice.

The Difference Amplifier Circuit

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- Example:
- a) Design a difference amplifier that amplifies the difference between two input voltages by a gain of 8, using an ideal op-amp and $\pm 8\ V$ power supplies.



- Using the simplified difference-amplifier equation:

$$v_o = \frac{R_b}{R_a}(v_b - v_a) = 8(v_b - v_a) \quad \text{so} \quad \frac{R_b}{R_a} = 8$$

We want two resistors whose ratio is 8. We can choose $R_b = 80\ k\Omega$ and $R_a = 10\ k\Omega$ although there are many other possibilities.

Note that simplified difference-amplifier equation requires that,

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

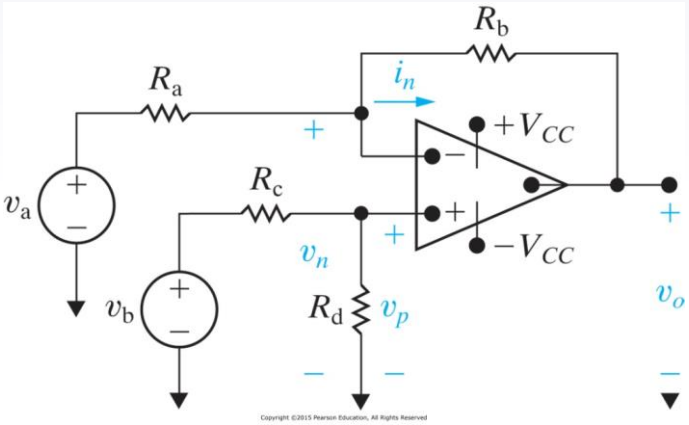
A simple choice would be $R_c = 15\ k\Omega$ and $R_d = 120\ k\Omega$

The Difference Amplifier Circuit

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- Example:

b) Suppose $v_a = 1\,V$ in the differenve amplifier designed in part (a). What range of input voltages for v_b can allow the op-amp to remain in its linear region?



- Solve two different version of the simplified difference-ampifier equation for v_o in terms of v_b, first using $v_0 = +8V$ and then using $v_0 = -8V$

$v_o = 8(v_b - 1) = 8\,V$ so $v_b = 2V$

$v_o = 8(v_b - 1) = -8\,V$ so $v_b = 0V$

Thus, if $v_a = 1\,V$ in the difference-amplifier from part (a), the op-amp will remain in its linear region of operation if $0 \leq v_b \leq +2V$

End of the Lesson (06.11.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist